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SYSTEM AND METHOD FOR SKIN TEMPERATURE REGULATION

Abstract

A system for regulating skin temperature includes an insulated unit including at least one opening for receiving an appendage of a user, a controller for setting a target internal temperature of the insulated unit, a temperature probe for measuring an actual internal temperature of the insulated unit, and a thermoelectric unit for providing at least one of heated and cooled air into the insulated unit for at least one of raising and lowering the actual internal temperature of the insulated unit. The controller is configured to control power and voltage provided to the thermoelectric unit to maintain the actual internal temperature within the target internal temperature range, such that the skin temperature of the appendage in the insulated unit is regulated within the target internal temperature range while providing medical treatment to the user.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to U.S. Provisional Application 63/334,385 filed on Apr. 25, 2022, the contents of which is incorporated herein by references in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates generally to an appliance for regulating and maintaining a constant skin temperature.

BACKGROUND

[0003] Every year, over 140,000 men, women, and children in the United States receive chemotherapy as a treatment for cancer. While chemotherapy is designed to kill fast-growing cancerous cells, healthy cells including nerves may often be damaged in the process. As a result, this damage may lead to neuropathy, damage to the sensory nerves in the hands and feet that can cause pain, tingling, and numbness. For many patients, neuropathy may be permanent and can prevent precision movements, return to work, and interfere with other activities of daily living.

[0004] Cooling of the hands and feet during chemotherapy administration, often referred to as icing, may prevent permanent damage in a large percentage of patients. Icing may be effective in preventing neuropathy by causing vasoconstriction or cold induced narrowing of the arteries to the skin and nerves. Some studies have also indicated a direct effect on the cellular metabolism, making the individual cells resistant to the side effects of medications. This may be similar to the cooling effect that can allow someone accidentally submerged in ice water to survive longer by slowing the metabolism.

[0005] Currently, icing may occur in a variety of methods. Some facilities or patients use ice packs, frozen bags, or submerge their hands and feet in ice water to experience the effect of icing. These methods may cause pain and are messy due to melting ice. The administration of chemotherapy may last up to six hours. As icing is typically done for the duration of chemotherapy, it may also cause frost bite in individuals. Frostbite can also cause neuropathy and can lead to additional tissue loss if severe.

[0006] Therefore, a need exists for an electrical cooling platform that provides a safe, painless experience to keep the digits within a therapeutic temperature range without the risk for frostbite. Additionally, a need exists for a system without the mess of ice or water. Hands and feet should be easily removed and slipped back into the device for better convenience and user experience during treatment, for example, including for trips to the restroom, to eat snacks, or to use mobile devices.

BRIEF DESCRIPTION

[0007] According to the disclosure, a system for regulating skin temperature includes an insulated unit including at least one opening for receiving an appendage of a user, a temperature probe for measuring an internal temperature of the insulated unit, a thermoelectric unit for providing at least one of heated and cooled air into the insulated unit; and a controller for maintaining the internal temperature of the insulated unit for regulating skin temperature of the appendage of the user such that the skin temperature is within a therapeutic range for providing medical treatment to the user.

[0008] Also according to the disclosure, a method of regulating skin temperature includes inserting an appendage of a user into at least one opening of an insulated unit, setting a target internal

temperature range of the insulated unit with a controller, measuring an actual internal temperature of the insulated unit with a temperature probe, providing at least one of heated and cooled air into the insulated unit via at least one thermoelectric unit and at least one fan to at least one of raise or lower the actual internal temperature, and controlling the power and voltage to the thermoelectric unit via the controller to maintain the actual internal temperature within the target internal temperature range.

[0009] According to the disclosure, a system for regulating skin temperature includes an insulated unit including at least one opening for receiving an appendage of a user, a controller for setting a target internal temperature of the insulated unit, a temperature probe for measuring an actual internal temperature of the insulated unit, and a thermoelectric unit for providing at least one of heated and cooled air into the insulated unit for at least one of raising and lowering the actual internal temperature of the insulated unit. The controller is configured to control power and voltage provided to the thermoelectric unit to maintain the actual internal temperature within the target internal temperature range, such that the skin temperature of the appendage in the insulated unit is regulated within the target internal temperature range while providing medical treatment to the user.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1A is a schematic view generally illustrating an outer view of an embodiment of a device used in connection with heating or cooling feet of a user.

[0011] FIG. 1B is a schematic view generally illustrating a cross sectional view of an embodiment of a device used in connection with heating or cooling feet of a user.

[0012] FIG. 1C is a schematic view generally illustrating the base of an embodiment of a device used in connection with heating or cooling feet of a user.

[0013] FIG. 1D is a schematic view generally illustrating a side view of two connected units of an embodiment of a device used in connection with heating or cooling feet of a user.

[0014] FIG. 2 is a schematic view generally illustrating an exemplary thermoelectric unit.

[0015] FIG. 3A is an external view generally illustrating a front view of an embodiment of a device used in connection with heating or cooling feet of a user.

[0016] FIG. 3B is an external view generally illustrating a side view of an embodiment of a device used in connection with heating or cooling feet of a user.

[0017] FIG. 3C is an external view generally illustrating a perspective view of an embodiment of a device used in connection with heating or cooling feet of a user.

[0018] FIG. 3D is an external view generally illustrating an embodiment of a control panel on a device used in connection with heating or cooling feet of a user.

[0019] FIG. 3E is an external view generally illustrating a perspective view of an embodiment of a device while in use by a user.

[0020] FIG. 4A is an internal view generally illustrating a side view of an embodiment of a device used in connection with heating or cooling feet of a user.

[0021] FIG. 4B is an internal view generally illustrating a perspective view of an embodiment of a device used in connection with heating or cooling feet of a user.

[0022] FIG. 5A is an external view generally illustrating a perspective view of an embodiment of a device used in connection with heating or cooling hands of a user with a viewing screen.

[0023] FIG. 5B is external view generally illustrating a perspective view of an embodiment of a device used in connection with heating or cooling hands of a user.

[0024] FIG. 6 is an exploded view generally illustrating an embodiment of a device used in connection with heating or cooling hands of a user.

[0025] FIG. 7 is an exemplary method of regulating skin temperature of a user with a device.

DESCRIPTION

[0026] Referring now to the discussion that follows and the drawings, illustrative approaches to the disclosed systems and methods are described in detail. Although the drawings represent some possible approaches, the drawings are not necessarily to scale and certain features may be exaggerated, removed, or partially sectioned to better illustrate and explain the present disclosure. Further, the descriptions set forth herein are not intended to be exhaustive, otherwise limit, or restrict the claims to the precise forms and configurations shown in the drawings and disclosed in the following detailed description.

[0027] This disclosure relates generally to a device that maintains a consistent skin temperature of a user's appendages. The device may be used while a user is actively partaking in chemotherapy treatment to prevent chemotherapy side-effects such as neuropathy and nail damage. The device is a small, portable, lightweight, and reusable electric appliance that creates cold air to cool the hands and feet of a user. The device may also be configured to provide warm air to heat the hands and feet of a user. An exemplary device includes a thermoelectric device such as a Peltier unit to create the cooled or heated air. The device may also include a temperature probe and feedback controller to regulate the temperature of the air. Using cold or warm air to regulate the temperature of the hands and feet of a user may maintain a constant skin temperature and provide therapeutic temperature to an appendage while the user receives medical treatment.

[0028] Units are built as thermoelectric driven, electrical units. The box is insulated. It is lined with a thin disposable fabric liner to ensure cleanliness and comfort. In an exemplary system, two oval openings provide a way to place appendages such as hands or feet comfortably within the unit. These openings are insulated with a soft sleeve with an elastic closure such as a cinch to prevent air from escaping the unit. When turned on, the heating or cooling rate of the unit is controlled and will not heat or cool more than 3.6° F. (2° C.) per minute to prevent pain to the user. The unit heat and cools to 60° F. with a range of plus or minus 5° F., keeping the skin temperature of the appendage in a therapeutic range for neuropathy prevention and avoiding temperatures known to cause frostbite with prolonged duration. The temperature of the skin is measured based on an external temperature of the skin on the appendage, however in maintaining the external temperature, the temperature throughout the layers of skin are also regulated. The unit includes a safety shut off below 50° F. and above 70° F. in order to prevent burning or freezing of the user's skin.

[0029] Referring now to the figures, FIGS. 1A-1D illustrate a system **100** for regulating skin temperature in an exemplary configuration for regulating temperature of feet of a user, however other configurations may be used for regulating temperature of different parts of the body, for example, hands of a user. The system **100** is a device **102** for regulating and maintaining a constant skin temperature. Device **102** includes an insulated unit **104** in which an appendage **110** is inserted into and in which temperature is cooled or heated and maintained. Device **102** is an electric device having an external power supply **106**, such as a power cord **112** and plug extending from a posterior end **108** of device **102**. Power cord **112** may be, in some instances, longer than eight feet long to ensure device **102** may be plugged into a power source behind a chair used in chemotherapy or other medical treatments. Insulated unit **104** may include a storage compartment **114**, or in instances, a wrapping mechanism, for management of power cord **112** to provide compact storage and portability when device **102** is not in use. Storage compartment **114** may be positioned on posterior end **108** of unit **104** in proximity to power supply **106**. In other examples, device **102** may be battery operated to provide therapeutic heating or cooling to a user where a external power supply is not available.

[0030] Insulated unit **104** may include a handle **116**, for example, on an anterior end **118**. Handle **116** is a u-shaped type for comfortability and ergonomic use by a user. Handle **116** may be of other shapes and sizes, but should remain comfortable for user who may be experiencing dry skin,

rashes, and pain from, for example, chemotherapy. Handle **116** may be foldable, for example, into a storage compartment, to increase compactness of insulated unit **104** for storage and travel.

[0031] In an exemplary configuration as illustrated in FIGS. **1A-1D**, insulated unit **104** includes openings **120** on a top surface **122** for insertion of a user's appendages **110**, for example feet, into insulated unit **104**. In exemplary configurations for regulating temperature of hands, for example, openings **120** may be located on a posterior end **108** for ease of appendage **110** insertion. Openings **110** are insulated, oval apertures in surface (e.g., top surface **122**, posterior end **108**) of insulated unit **104** for providing a location to place targeted appendage **110** comfortably into device **102**. Openings **110** may include a sleeve **124** with a closing device **126** such as an elastic closure or a cinch. Sleeve **124** provides comfort to a user to ensure appendage **110** is in contact with a soft surface rather than a hard exterior cover of unit **104**. Closing device **126** tightens sleeve **124** around appendage **110** to prevent air from escaping from insulated unit **104** while in use and the closing device **126** is pulled tight. Openings **110** are further lined with a disposable liner **128** configured to provide comfort and further insulation to user, as well as to ensure sanitation amongst multiple users of device as liners **128** may be replaced between device **102** uses.

[0032] Device **102** includes a control panel **130** for use by a user or, for example, a medical practitioner providing medical service to the user. In the exemplary configuration of FIGS. **1A-1D**, control panel **130** is located on a top surface **120** for ease of use by a user bending down to operate device **102** while feet are positioned in openings **110**. In other examples, control panel **130** may be located on other surfaces (e.g., posterior end **108** or a side surface). Control panel **130** includes buttons **132** for, for example, turning power on to device **102** and for controlling temperature. Control panel **130** may include additional operations including, but not limited to, USB power ports **134** for connecting Bluetooth connection devices such as touchpads or a mouse for use with a laptop, cell phone, or other external electronic devices. Control panel cord **136** connects control panel **130** to a controller **138** for operation of device **102**.

[0033] Positioned inside insulated unit **104** and towards a bottom surface **140** is a thermoelectric unit **142**, which in examples, may be a Peltier unit. An exemplary thermoelectric unit is illustrated in FIG. **2**. Thermoelectric unit **142** includes two ceramic plates **202**, **204**, separated by semiconductor pellets **206**, **208**. A first ceramic plate **202** absorbs heat, while a second ceramic plate **204** dissipates heat when a current is passed through a first semiconductor pellet **206** and a second semiconductor pellet **208**. In an embodiment of device **2**, current is passed to first ceramic plate **202** that absorbs heat such that thermoelectric unit **142** is a cooling unit. In another embodiment, current direction may be modified and reversed such that current is provided to second ceramic plate **204** that dissipates heat such that thermoelectric unit **142** is a heating unit. Thermoelectric unit **142** includes at least one set of ceramic plates **202**, **204** and semiconductor pellets **206**, **208**, but may include additional sets for increased heating and cooling power.

[0034] Thermoelectric unit **142** includes an at least first fan **144**. First fan **144** is driven by electricity to cycle cooled air from thermoelectric unit **142** into and throughout insulated unit **142**. Fan **144** runs continuously while device **102** is powered to provide a white noise to user to not disturb users and individuals within a close proximity to device **102** and to ensure rotation of air inside device **102** and accurate temperature control. For example, fan **144** mixes air throughout device **102** such that the internal temperature remains consistent throughout device **102** and hot spots or cool spots are prevented and/or minimized. Thermoelectric unit **142** is positioned adjacent to controller **138** and is provided with information from controller **138** to determine when to operate thermoelectric unit **142** and in which operation (e.g., heating operation or cooling operation) thermoelectric unit **142** should operate in to stay in a target temperature range. A temperature probe **146** is connected to controller **138** for measuring and monitoring an internal temperature of insulated box **104**. Temperature probe **146** is configured to work in conjunction with controller **138**, communicating the internal temperature to controller **138** such that controller **138** can provide operational controls to thermoelectric unit **142** to produce air of a necessary

temperature to maintain target temperature range. Temperature probe **146** may additionally include a skin temperature probe for measuring an external skin temperature of the user compared to internal temperature of insulated unit **104**.

[0035] Controller **138** and thermoelectric unit **142** are configured to operate in conjunction with each other such that thermoelectric unit **142** operates based on information received from controller **138**. In an exemplary configuration, thermoelectric unit **142** is configured to heat or cool at a controlled rate and will not heat or cool more than 3.6° F. per minute to prevent pain to a user from rapid cooling and/or heating. In an exemplary setting, thermoelectric unit **142** cools or heats to approximately 60° F., measured by temperature probe **146**. Thermoelectric unit **142** may heat or cool to a range of plus or minus 5° F. from 60° F. to keep appendage **110** temperature within a target therapeutic range for neuropathy prevention and avoiding temperatures known to cause frostbite with prolonged duration. Controller **138** includes a safety shut off **148** configured to turn power to thermoelectric unit **142** off when temperature probe **146** measures below 50° F. or above 70° F.

[0036] Illustrated in FIG. 1C is an exemplary configuration of device **102** viewed from bottom surface **140**. Device **102** includes thermoelectric unit **142** and controller **138** located proximate the bottom of insulated unit **104**. Extending outward from insulated unit **104** on posterior end **108** is power supply **106**, **108** for providing power to device **102**. Extending from controller **138** and located in insulated unit **104** is temperature probe **146** for measuring temperature inside insulated unit **104**. Extending from controller **138** and built into walls of insulated unit **104** is control panel cord **136** for controlling controller **138** based on received inputs from control panel **130**.

[0037] Illustrated in FIG. 1D is an exemplary configuration of a first device **102A** and a second device **102B** connected to each other for portability of more than one device **102** and viewed from a side perspective. In the exemplary configuration, first device **102A** and second device **102B** are placed adjacent to each other at their bottom surfaces **140**. A first strap **148** may connect first device **102A** to second device **102B** at a posterior end **108** of each of the devices. For example, first strap **148** may be connected to each of first device **102A** and second device **102B** via connector **154**, for example, a snap fastener or a buckle. A second strap **150** may connect first device **102A** to second device **102B** at an anterior end **118** of each device. For example, second strap **150** may be connected to each of first device **102A** and second device **102B** via connector **154**. Connecting first device **102A** and second device **102B** is provided for ease of a user's ability to travel with multiple devices **102**. On anterior end **118**, a handle strap **152** may attach first handle **116A** from first device **102A** to second handle **116B** from second device **102B** so a user may conveniently hold handles **116A**, **116B** from both devices **102A**, **102B** concurrently.

[0038] Illustrated in FIGS. 3A-3E is an exemplary embodiment of device **102** configured to be used for regulating and maintaining temperature of feet of a user. FIG. 3A illustrates device **102** from anterior end **118**. Insulated box **104** is surrounded by an external cover **302** that provides protection and concealment of the internal electronics. External cover **302** may include a venting grill **304**, for example, on anterior end **118** of device **102**. Grill **304** is designed to expel air from within insulated unit **104** such as a warm air exhaust.

[0039] Referring to FIG. 3B, an exemplary embodiment of device **102** is illustrated from a side view. Insulated unit **104** includes a user engaging portion **306** at posterior end **108** and a covered portion **308** at anterior end **118**. User engaging portion **306** includes openings **120** for a user to insert appendages **110** into insulated unit **104**. In embodiments, user engaging portion **306** may be angled at a slightly upward angle for user comfortability (i.e., for a user to insert feet into openings **120** while sitting in a chair). In additional embodiments, user engaging portion **306** may be relatively flat for user comfortability (i.e., for a user to slide hands into openings **120** when sitting at a table).

[0040] Referring to FIG. 3C, an exemplary embodiment of device **102** is illustrated from a perspective view, generally facing a first side and posterior end **108**. User engaging portion **306**

includes a raised protrusion **310** from external cover **302** to provide openings **110** at an upward angle to user. FIG. **3D** illustrates an exemplary embodiment of device **102** illustrated from a perspective view, generally facing a second side and posterior end **108**. In embodiments, control panel **130** may be positioned towards posterior end **108** of device **102**. Control panel **130** includes buttons **132** for user to input feedback and controls regarding, for example, power and temperature selection. Additionally, control panel **130** may include, for example, USB ports **134** and auxiliary cord ports **312** for connection and use of Bluetooth devices such as computer mice or touchpads. FIG. **3E** illustrated an exemplary embodiment of device **102** illustrated from a perspective view, generally facing a first side and posterior end **108**, while in use by a user. User is illustrated sitting with appendages **110** (i.e., feet) inserted into openings **120** of insulated unit **104**.

[0041] As illustrated in FIGS. **4A** and **4B**, an exemplary configuration of device **102** is illustrated from an internal view. Openings **120** include sleeve **124** around outer rim of opening **120** to provide comfort to appendage **110** when inserted in opening **120**. Openings **120** provide access to insulated bag **408** for appendage **110**. Insulated bags **408** may include an angled portion **406** which connects insulated bag **408** to opening **120** in configurations with raised protrusion **310** (i.e., covering portion of a user's ankle and legs when sitting with feet in bags **408**). Bags **408** may be made of a substantially thin fabric material or non-weaving fabric using a cut-and-sew process. A foam outer later (as illustrated in FIG. **6**) provides insulation to bag **408**. Insulated bags may additionally be lined internally with liner **128** (e.g., a disposable liner) for sanitation between uses such as where devices **102** are rented or used in a facility with multiple patients.

[0042] Referring to FIG. **4A**, device **102** illustrates a side view of the internal components of device **102**. Electricity received from power supply **106** is provided to a power adapter **402** and power boards **404** for operating device **102**. Power adapter **402** and power boards **404** may be positioned adjacent bottom surface **140** of device **102** to position weight of power adapter **402** and power board **404** towards a lower surface. Power adapter **402** and power board **404** are used to provide power to and operate controller **138**, temperature probe **146**, and thermoelectric unit **142** based on input from control panel **130**.

[0043] Referring to FIG. **4B**, device **102** illustrates a perspective view, generally from a side and anterior end, of the internal components of device **102**. Thermoelectric unit **142** may be positioned toward anterior end **118** of insulated unit **104** to save space in a Y-dimension of insulated unit **104**, as well as being positioned close to venting grill **304** to allow venting of exhaust waste. Control panel **130** may be positioned on a side surface toward posterior end **108** for easy access to a user.

[0044] An exemplary configuration of device **102** configured to regulate or maintain temperature of a user's hands is illustrated in FIGS. **5A** and **5B**. In embodiments, as illustrated in FIG. **5A**, device **102** may include a generally flat top surface **122**. In embodiments, as illustrated in FIG. **5B**, device **102** may include a raised protrusion **310** on top surface **122**. As openings **120** are below raised protrusion **310** in the illustrated embodiment, raised protrusion **310** is provided as an angled surface for facing a user's face (i.e., for placing reading or electronic materials for user viewing).

[0045] As illustrated in FIG. **5A**, device **102** has a generally flat top surface **122**. User engaging portion **306** is at posterior end **108**, such that openings **120** are generally on posterior end **108** facing surface. Openings **120** are configured such that device **102** may be placed on a surface, for example a table, and a user can slide their appendage (i.e., their hands) into openings **120** and lay their arms flat on the surface and/or table. As illustrated in FIG. **5B**, device **102** may include raised portion **310** on top surface **122** of external cover **302**. In embodiments as illustrated in FIG. **5B**, openings **120** are generally on posterior end **108** facing surface in a substantially flat configuration. Raised portion **310** is positioned above openings **120** such that raised portion **310** may provide a viewing surface **510** to a user. Viewing surface **510** may be generally angled upward toward a user's face such that viewing materials, for example a book or connected device **504**, may rest on viewing surface and be visible to a user to engage with while using device **102**. A ledge **506** is provided near a bottom of viewing surface **502** to provide a surface for viewing materials **504** to

rest on when positioned on viewing surface **502**.

[0046] FIG. **6** illustrates an exploded view of an embodiment of device **102** configured for cooling or heating appendages (i.e., hands) of a user. Insulated unit **104** includes several layers to ensure proper insulation for higher effectiveness due to loss of cool and/or hot air and temperature maintenance. External cover **302** of insulated box **104** includes a top cover **602** and a bottom cover **614** nestled together such that top cover **602** and bottom cover **614** make up substantially top surface **122**, bottom surface **140**, and both side surfaces of external cover **302**. A back cover **608** is provided to make up substantially a posterior end **108** facing surface of external cover **302**. Venting grill **304** is encompassed in a grill frame **610**. The combined venting grill **304** in grill frame **610** makes up anterior end **118** facing surface of external cover.

[0047] Within external cover **310** components, a foam layer **604** is generally positioned adjacent to external cover **310** in an oval tube shape. Foam layer **604** is provided to act as a source of insulation to maintain temperature within insulated box **104** to reduce energy production and increase effectiveness of device **102**. Located inward of foam layer **604** is an inner frame **606** to provide structure to internal components of device **102** such as the thermoelectric unit **142** and insulated bags **408**. Insulated bags **408** are positioned inside inner frame **606**. On an anterior end **118** facing side of inner frame, a thermoelectric unit frame **614** is included to house the at least one thermoelectric unit **142** and the at least one fan **144**. Venting grill **304** and grill frame **610** are positioned generally anterior to inner frame **606** and thermoelectric unit **138**. Power adapter **402** and power boards **404** are positioned generally adjacent to bottom cover **402** to distribute power assembly weight towards bottom surface **140**. Back cover **608** includes viewing surface **502** and ledge **506** in illustrated embodiment. Openings **120** are included in back cover **120**. Sleeves **124** are included in insulated bags **408** and extend out of openings **120** to provide comfortable edge to openings **120** for user.

[0048] Referring now to FIG. **7**, an exemplary method of operating heating and/or cooling device **102** to regulate skin temperature of a user is provided. At **700**, user begins operating device **102**. At **702**, user inserts an appendage **110** into insulated unit **104**, for example, inserting hands or feet of the user into openings **120** and insulated bags **408** of insulated unit **104**. At **704**, user operates control panel **130** to power device **102** on and set a target temperature range using buttons **132** on control panel **130**. Device **102** may also be Bluetooth enabled and could receive input for power or target temperature range via Bluetooth capable devices. A target temperature range may be, for example, between 55° F. to 65° F. Target temperature range of 55° F. to 65° F. is ideal for keeping appendage temperature in a therapeutic range for neuropathy prevention and avoiding temperatures prone to cause frostbite at prolonged duration. At **706**, temperature probe **146** measures an actual internal temperature of insulated unit **104**. In embodiments with a skin temperature probe, skin temperature probe may measure actual external skin temperature. At **708**, controller **138** compares actual internal temperature and/or actual external skin temperature to the target temperature range. If actual internal temperature and/or actual external skin temperature is not within target temperature range, controller **138** provides power to thermoelectric unit **142** to activate the at least one thermoelectric unit **142** and at least one fan **144**. Power and voltage are provided to thermoelectric unit **142** to provide heated or cooled air to insulated unit **104**. For example, if actual temperature is above target temperature range, voltage will be provided to thermoelectric unit **142** to provide cooled air to insulated unit **104** to decrease actual temperature. If actual temperature is below target temperature unit, voltage is reversed and thermoelectric unit **142** will provide heated air to insulated unit **104** to raise actual temperature. Controller **138** controls thermoelectric unit **142** to ensure actual temperature is not increased or decreased by a change of more than 3.9° F. per minute to prevent pain for fast cooling or heating to skin on appendage **110**. At **710**, controller **138** continues to maintain control of thermoelectric unit **142** by increasing or reducing power and/or reversing voltage to thermoelectric unit **142** to maintain the actual temperature within the target temperature range. Controller **138** will continue to maintain temperature until **712**, when user ends

operation of device **102** by shutting device **102** off or when a safety shut off pulls power from device **102** if actual temperature reaches an unsafe lever, for example, below 50° F. or above 70° F. [0049] The disclosed technology may be configured into a therapy helmet for preventing hair loss during chemotherapy. For example, a device **102** may be configured into a helmet design similar to a free-standing hair dryer for providing controlled heating or cooling of the scalp of a user during administration of chemotherapeutic agents that cause hair loss. A helmet design utilizes continuous skin temperature monitoring to assure maintenance of therapeutic temperature range and frostbite protection. This scalp device utilizes circulating air and continuous skin temperature monitoring to assure efficacy and prevent skin damage.

[0050] The disclosed technology may also be configured as a heating or cooling device for physical therapy, postoperative cooling, and for treatment of sports injuries. For example, the device **102** may be configured to include insulated bags **408** which conform to body contours and provides controlled heating or cooling, or heating alternative with cooling, for a variety of skeletal or muscular therapies. Skin temperature feedback from temperature probe **146** will assure the device is operating in a therapeutic range and maximum and minimum temperature controls will prevent burns or frost bite to vulnerable tissues. The disclosed device utilizes circulating air and continuous skin temperature monitoring to assure efficacy and prevent skin damage.

[0051] The disclosed technology may be modified to be used as a device for sports activities involving hands and feet. The device can be used as designed for controlled cooling after sports activities causing swelling in hands and feet, for example marathon or ultra-runners that treat swelling and contact with hot pavement which can be severe causing blistering, skin loss, and inability to wear shoes. The device utilizes circulating air and continuous skin temperature monitoring to assure efficacy and prevent skin damage.

[0052] The disclosed technology may be modified to be used as a heating device for sports activities, hypothermia, Reynaud's Syndrome, or neuropathy. This device can be used as designed as a controlled device for hands and feet by reversing the voltage of the thermoelectric unit to provide warm air. The device temperature control range can measure the skin temperature and adjust to warm the hands and feet by 1-2° F. per minute prevent pain from rewarming. Skin temperature of the hands and feet is usually 80° F.-84° F. at a room temperature of 70° F. Maximum temperature can be set at 105° F. to prevent burns. This application can be used for symptomatic relief of muscle spasm, Reynaud's Syndrome, and neuropathy. Other uses would include rewarming after cold climate sports activities such as skiing, ice skating, hockey, or after industrial work in cold environments. The device may also be used to provide controlled rewarming of hands and feet as a treatment for hypothermia. The device utilizes circulating air and continuous skin temperature monitoring to assure efficacy and prevent skin damage.

[0053] Thus, according to the disclosure, a system for regulating skin temperature includes an insulated unit including at least one opening for receiving an appendage of a user, a temperature probe for measuring an internal temperature of the insulated unit, a thermoelectric unit for providing at least one of heated and cooled air into the insulated unit; and a controller for maintaining the internal temperature of the insulated unit for regulating skin temperature of the appendage of the user such that the skin temperature is within a therapeutic range for providing medical treatment to the user.

[0054] Also according to the disclosure, a method of regulating skin temperature includes inserting an appendage of a user into at least one opening of an insulated unit, setting a target internal temperature range of the insulated unit with a controller, measuring an actual internal temperature of the insulated unit with a temperature probe, providing at least one of heated and cooled air into the insulated unit via at least one thermoelectric unit and at least one fan to at least one of raise or lower the actual internal temperature, and controlling the power and voltage to the thermoelectric unit via the controller to maintain the actual internal temperature within the target internal temperature range.

[0055] According to the disclosure, a system for regulating skin temperature includes an insulated unit including at least one opening for receiving an appendage of a user, a controller for setting a target internal temperature of the insulated unit, a temperature probe for measuring an actual internal temperature of the insulated unit, and a thermoelectric unit for providing at least one of heated and cooled air into the insulated unit for at least one of raising and lowering the actual internal temperature of the insulated unit. The controller is configured to control power and voltage provided to the thermoelectric unit to maintain the actual internal temperature within the target internal temperature range, such that the skin temperature of the appendage in the insulated unit is regulated within the target internal temperature range while providing medical treatment to the user.

[0056] When introducing elements of various embodiments of the disclosed materials, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Furthermore, any numerical examples in the following discussion are intended to be non-limiting, and thus additional numerical values, ranges, and percentages are within the scope of the disclosed embodiments.

[0057] While the preceding discussion is generally provided in the context of a heating or cooling device to be used on hands or feet of a chemotherapy patient, it should be appreciated that the present techniques are not limited to such limited contexts. The provision of examples and explanations in such a context is to facilitate explanation by providing instances of implementations and applications. The disclosed approaches may also be utilized in other contexts or configurations such as cold therapy helmets. The device may also be reconfigured to a different temperature range, providing heating or cooling for treatments such as for physical therapy, postoperative cooling, or treatment from sports injuries.

[0058] While the disclosed materials have been described in detail in connection with only a limited number of embodiments, it should be readily understood that the embodiments are not limited to such disclosed embodiments. Rather, that disclosed can be modified to incorporate any number of variations, alterations, substitutions, or equivalent arrangements not heretofore described, but which are commensurate with the spirit and scope of the disclosed materials. Additionally, while various embodiments have been described, it is to be understood that disclosed aspects may include only some of the described embodiments. Accordingly, that disclosed is not to be seen as limited by the foregoing description but is only limited by the scope of the appended claims.

Claims

1. A system for regulating skin temperature, comprising: an insulated unit including at least one opening for receiving an appendage of a user; a temperature probe for measuring an internal temperature of the insulated unit; a thermoelectric unit for providing at least one of heated and cooled air into the insulated unit; and a controller for maintaining the internal temperature of the insulated unit for regulating the skin temperature of the appendage of the user such that the skin temperature is within a therapeutic range for providing medical treatment to the user.
2. The system of claim 1, wherein the controller maintains the internal temperature of the insulated unit within a target internal temperature range between 55° F. to 65° F.
3. The system of claim 1, further including a safety shut off for stopping power to the thermoelectric unit when the internal temperature is one of below 50° F. and above 70° F.
4. The system of claim 1, wherein the thermoelectric unit is configured to provide the at least one of heated and cooled air into the insulated unit such that the internal temperature of the insulated unit does not exceed a change of 2° F. per minute.
5. The system of claim 1, wherein the at least one opening is positioned substantially toward a

posterior end of the insulated unit and is configured to receive at least one of hands or feet of the user.

6. The system of claim 5, further including a sleeve surrounding an edge of the at least one opening, the sleeve including a cinch to secure the sleeve to the appendage of the user.

7. The system of claim 1, further including a control panel, the control panel configured to receive input from the user for providing power to the insulated unit and controlling the internal temperature.

8. The system of claim 1, further including at least one fan for cycling air throughout insulated unit for ensuring the internal temperature is substantially consistent throughout the insulated unit.

9. The system of claim 1, wherein the temperature probe includes a skin temperature probe for measuring the external skin temperature of the appendage.

10. The system of claim 1, further comprising a venting grill for expelling exhaust air from insulated unit to maintain the internal temperature of insulated unit.

11. The system of claim 1, further comprising an external cover surrounding the insulated unit; and wherein the external cover further includes a raised protrusion generally towards the posterior end of the insulated unit.

12. The system of claim 11, wherein the raised protrusion includes a viewing surface and a ledge.

13. The system of claim 11, wherein the raised protrusion includes the at least one opening.

14. The system of claim 1, further including an insulated bag attached to the at least one opening to surround the appendage in the insulated unit.

15. The system of claim 14, further including a disposable liner to line an inner surface of the insulated bag.

16. A method of regulating skin temperature, comprising: inserting an appendage of a user into an opening of an insulated unit; setting a target internal temperature range of the insulated unit with a controller; measuring an actual internal temperature of the insulated unit with a temperature probe; providing at least one of heated and cooled air into the insulated unit via at least one thermoelectric unit and at least one fan to at least one of raise or lower the actual internal temperature; and controlling the power and voltage to the thermoelectric unit via the controller to maintain the actual internal temperature within the target internal temperature range to regulate skin temperature of the appendage to provide medical treatment to the appendage.

17. The method of claim 16, further comprising maintaining the target internal temperature range between 55° F. to 65° F.

18. The method of claim 16, further comprising at least one of raising or lowering the actual internal temperature no more than 2° F. per minute.

19. The method of claim 16, further comprising shutting off power to the thermoelectric unit if the actual internal temperature is at least one of below 50° F. and above 70° F.

20. A system for regulating skin temperature, comprising: an insulated unit including at least one opening for receiving an appendage of a user; a controller for setting a target internal temperature range of the insulated unit; a temperature probe for measuring an actual internal temperature of the insulated unit; and a thermoelectric unit for providing at least one of heated and cooled air into the insulated unit for at least one of raising and lowering the actual internal temperature of the insulated unit; wherein the controller is configured to control power and voltage provided to the thermoelectric unit to maintain the actual internal temperature within the target internal temperature range, such that the skin temperature of the appendage in the insulated unit is regulated within the target internal temperature range while providing medical treatment to the user.
